

Class 13: Core discussion networks and loneliness

Matthew J. Salganik

Sociology 204: Social Networks
Princeton University

March 17, 2025

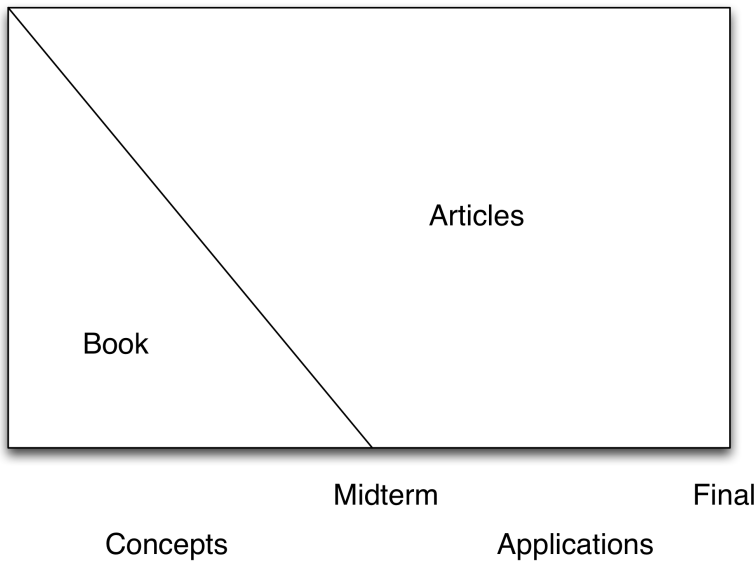


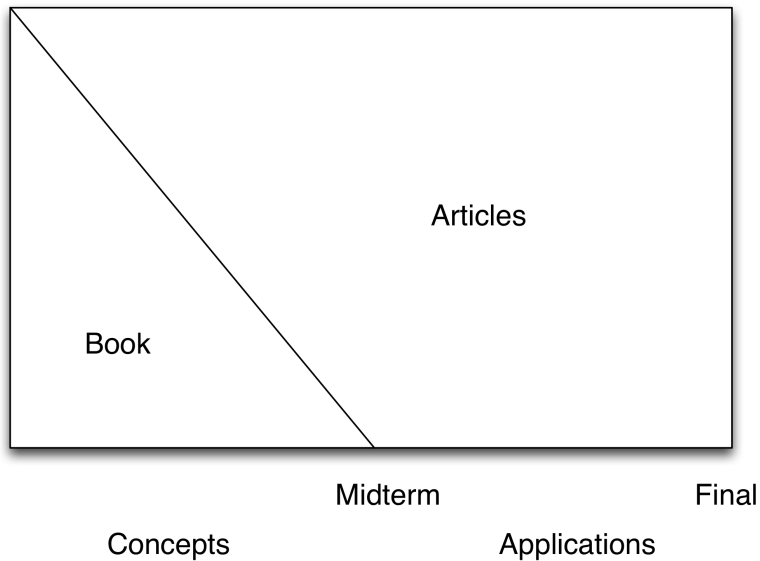
Welcome back from spring break!

Welcome back from spring break!

Reminders

- ▶ you can submit midterm corrections by Wednesday at 11am. This is optional.
There is no other assignment this week.
- ▶ there is a quiz and precept this week.





My guess is that the pre-reads will become more important.

- ▶ Students will be able to *describe* foundational ideas used in the study of networks.
- ▶ Students will be able to *describe* the interconnections between these ideas.
- ▶ Students will be able to *use* these ideas to gain insight into real-world phenomena.
- ▶ Students will be able to *evaluate* research that applies these ideas to real-world phenomena.
- ▶ Students will be able to *create* research about real-world phenomena.

New laptop policy

- ▶ Default: No laptops in lecture

New laptop policy

- ▶ Default: No laptops in lecture
- ▶ If you would like to use a laptop in lecture, please email me explaining your reasons so that we can learn from them. All reasonable requests will be accepted.

New laptop policy

- ▶ Default: No laptops in lecture
- ▶ If you would like to use a laptop in lecture, please email me explaining your reasons so that we can learn from them. All reasonable requests will be accepted.
- ▶ All students using laptops must sit in the last row so as not to distract their classmates (again email me if you need a change).

New laptop policy

- ▶ Default: No laptops in lecture
- ▶ If you would like to use a laptop in lecture, please email me explaining your reasons so that we can learn from them. All reasonable requests will be accepted.
- ▶ All students using laptops must sit in the last row so as not to distract their classmates (again email me if you need a change).
- ▶ Questions?

New laptop policy

- ▶ Default: No laptops in lecture
- ▶ If you would like to use a laptop in lecture, please email me explaining your reasons so that we can learn from them. All reasonable requests will be accepted.
- ▶ All students using laptops must sit in the last row so as not to distract their classmates (again email me if you need a change).
- ▶ Questions?

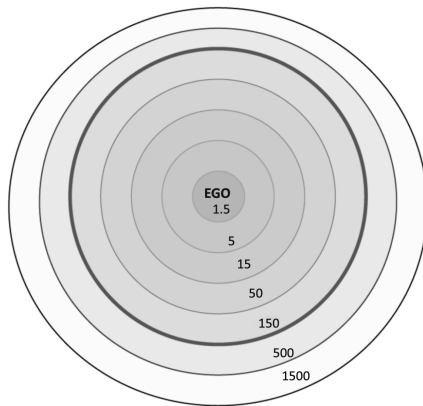
More information:

<https://www.nytimes.com/2017/11/22/business/laptops-not-during-lecture-or-meeting.html> |

<https://socialscience.msu.edu/news-events/news/archives/2021/2021-06-01.html>

<https://mcgraw.princeton.edu/guiding-students-using-technology-appropriately-suggestions-your-syllabus>

1. Marsden (1985). Core discussion networks of Americans. *American Sociological Review*, 52(1):122-131.
2. Lee et al. (2023). Transformation of social relationships in COVID-19 America: Remote communication may amplify political echo chambers. *Science Advances* 9(511):1-14.
3. Murthey (2023). Surgeon General: We Have Become a Lonely Nation. It's Time to Fix That. *New York Times*.



Trends in Cognitive Sciences

Figure 2. The Circles of Friendship. An individual (ego) sits within a series of hierarchically inclusive circles, or layers, of friendship that have a very distinct scaling ratio (each layer is three times the size of the one inside it). The heavy line demarcates the 150 layer, the typical size of personal social networks. Beyond this are acquaintances (500 layer); the layer at 1500 appears to represent the limit on the number of faces we can put names to. Frequency of contact, rated emotional closeness, and willingness to act altruistically towards a given alter all decline across the successive circles [2,10].

What do the core discussion networks of Americans look like?

- ▶ How big are they?
- ▶ How homogenous are they?
- ▶ How do these vary by respondent's characteristics such as age, sex, race/ethnicity, geographic location?

NETWORK ITEMS AND THE GENERAL SOCIAL SURVEY *

Ronald S. BURT **

Columbia University

This is an argument for obtaining network data in the General Social Survey (GSS). The proposal requires a discussion of how and why at least minimal network data ought to be obtained in a probability sample survey of attitudes and behaviors.

I begin with general concerns; briefly describing the proposal, available experience with the proposed items in large probability samples, how the proposed items are different from existing GSS items, kinds of variables that the proposed items would generate, and kinds of research questions that could be addressed if the proposed items were included in the GSS.

I then address comparatively focused questions likely to arise in deliberations over the proposal; explaining how much interview time the proposed items are expected to require, why one rather than multiple name generators are proposed, why recording five alters is proposed, why intimacy is proposed as the name generator criterion content, why a short form is proposed for obtaining formal data, how priorities among name interpreter attribute items were established, how the proposed items elicit data on the strength and content of relationships, and how the proposed data might be coded for easy access by GSS users.

[https://doi.org/10.1016/0378-8733\(84\)90007-8](https://doi.org/10.1016/0378-8733(84)90007-8)

Three main kinds of questions:

- ▶ Name generator: “From time to time, most people discuss important matters with other people. Looking back over the last six months – who are the people with whom you discuss matters important to you? Just tell me their first names or initials.”

Three main kinds of questions:

- ▶ Name generator: “From time to time, most people discuss important matters with other people. Looking back over the last six months – who are the people with whom you discuss matters important to you? Just tell me their first names or initials.”
- ▶ Name interpreter (asked about each alter, many possible questions): What is the age of MJ?

Three main kinds of questions:

- ▶ Name generator: “From time to time, most people discuss important matters with other people. Looking back over the last six months – who are the people with whom you discuss matters important to you? Just tell me their first names or initials.”
- ▶ Name interpreter (asked about each alter, many possible questions): What is the age of MJ?
- ▶ Relationship interpreter (asked about each pair of alters): Tell me about the relationship between MJ and DT? Would you say they are especially close, total strangers, or somewhere in between?

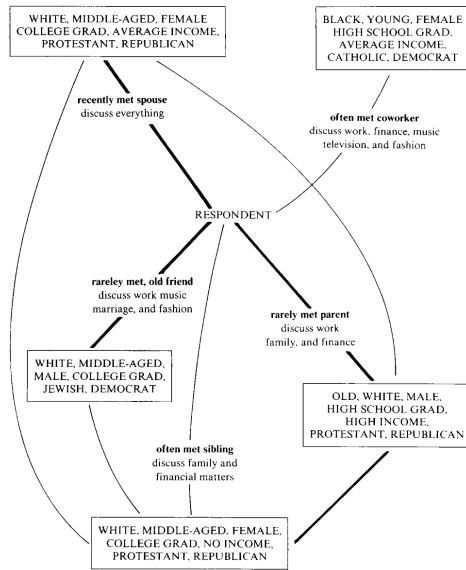
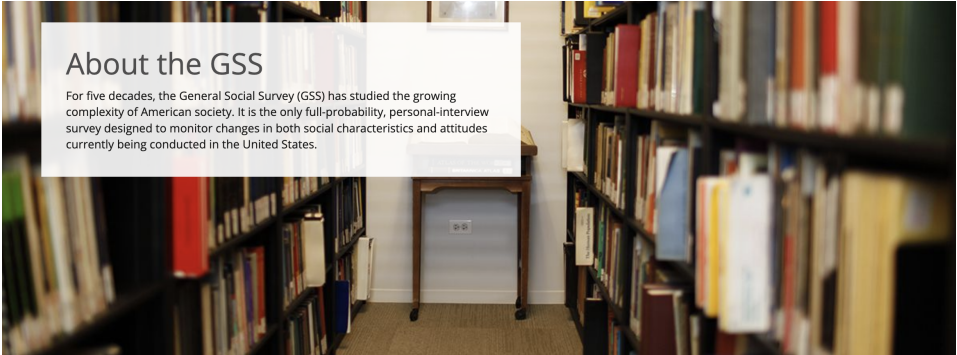


Figure 2. Formal data in Figure 1 enriched with some of the proposed name interpreter data (Q5 through Q15) on the hypothetical respondent and alters.



About the GSS

For five decades, the General Social Survey (GSS) has studied the growing complexity of American society. It is the only full-probability, personal-interview survey designed to monitor changes in both social characteristics and attitudes currently being conducted in the United States.

You can download this data and use it for your own research. More information:
<https://gss.norc.org/us/en/gss/about-the-gss.html>

CORE DISCUSSION NETWORKS OF AMERICANS*

PETER V. MARSDEN

University of North Carolina at Chapel Hill

Aspects of interpersonal networks in which Americans discuss "important matters" are examined using data from the 1985 General Social Survey. These are the first survey network data representative of the American population. The networks are small, kin-centered, relatively dense, and homogeneous in comparison with the sample of respondents. Bivariate examination of subgroup differences by age, education, race/ethnicity, sex, and size of place indicates that network range is greatest among the young, the highly educated, and metropolitan residents. Sex differences consist primarily of differences in kin/nonkin composition of networks.

<https://www.jstor.org/stable/2095397>

Table 1. Univariate Distributions of Measures of Network Form and Composition

Variable	Value	%	Mean	Standard Deviation	(N)
Overall network size	0	8.9	3.01	1.77	1531
	1	14.9			
	2	15.3			
	3	21.0			
	4	15.2			
	5	19.2			
	6+	5.5			
Kin network size ^a	0	26.4	1.53	1.34	1531
	1	29.6			
	2	21.8			
	3	12.6			
	4	6.3			
	5	3.3			
Nonkin network size ^a	0	36.4	1.40	1.41	1531
	1	22.2			
	2	18.9			
	3	13.0			
	4	6.3			
	5	3.1			
Proportion kin	0.00	19.2	0.55	0.37	1395
	0.01-0.33	15.4			
	0.34-0.66	20.7			
	0.67-0.99	14.5			
	1.00	30.2			
Density	<.25	8.1	0.61	0.28	1161
	0.25-0.49	18.0			
	0.50-0.74	39.5			
	>0.74	34.4			
Age heterogeneity (std. deviation)	<5	25.0	10.54	6.39	1153
	5-<10	23.2			
	10-<15	25.7			
	15+	26.1			
Population heterogeneity			17.91		
Education heterogeneity (std. deviation)	0-1	29.8	1.78	1.37	1132
	>1-2.5	43.3			
	>2.5	26.9			
Population heterogeneity			3.17		
Race heterogeneity (IQV)	0	91.8	0.05	0.18	1167
	>0	8.2			
Population heterogeneity			0.38		
Sex heterogeneity (IQV)	0	22.4	0.68	0.38	1167
	0.01-0.90	40.7			
	>0.90	36.9			
Population heterogeneity			0.99		

^a Information about kinship was collected on only the first five alters cited. For this reason the sum of the distributions for kin network size and nonkin network size is not identical to the distribution of overall network size.

Overall network size	0	8.9	3.01	1.77	1531
	1	14.9			
	2	15.3			
	3	21.0			
	4	15.2			
	5	19.2			
	6+	5.5			

mean of 3, 9% report 0.

Kin network size ^a	0	26.4	1.53	1.34	1531
	1	29.6			
	2	21.8			
	3	12.6			
	4	6.3			
	5	3.3			
Nonkin network size ^a	0	36.4	1.40	1.41	1531
	1	22.2			
	2	18.9			
	3	13.0			
	4	6.3			
	5	3.1			
Proportion kin	0.00	19.2	0.55	0.37	1395
	0.01–0.33	15.4			
	0.34–0.66	20.7			
	0.67–0.99	14.5			
	1.00	30.2			

About 30% only report discussing important matters with kin

Density	<.25	8.1	0.61	0.28	1161
	0.25–0.49	18.0			
	0.50–0.74	39.5			
	>0.74	34.4			

For many people, the networks are dense (similar to high clustering coefficient from the small world models)

These results vary by age, level of education, race/ethnicity, sex, and size of place.

- ▶ greatest difference by age and level of education
- ▶ appreciable difference by race/ethnicity and size of place
- ▶ small difference by gender

What has happened since 1985?

Social Isolation in America: Changes in Core Discussion Networks over Two Decades

Miller McPherson

University of Arizona and Duke University

Lynn Smith-Lovin

Duke University

Matthew E. Brashears

University of Arizona

Have the core discussion networks of Americans changed in the past two decades? In 1985, the General Social Survey (GSS) collected the first nationally representative data on the confidants with whom Americans discuss important matters. In the 2004 GSS the authors replicated those questions to assess social change in core network structures. Discussion networks are smaller in 2004 than in 1985. The number of people saying there is no one with whom they discuss important matters nearly tripled. The mean network size decreases by about a third (one confidant), from 2.94 in 1985 to 2.08 in 2004. The modal respondent now reports having no confidant; the modal respondent in 1985 had three confidants. Both kin and non-kin confidants were lost in the past two decades, but the greater decrease of non-kin ties leads to more confidant networks centered on spouses and parents, with fewer contacts through voluntary associations and neighborhoods. Most people have densely interconnected confidants similar to them. Some changes reflect the changing demographics of the U.S. population. Educational heterogeneity of social ties has decreased, racial heterogeneity has increased. The data may overestimate the number of social isolates, but these shrinking networks reflect an important social change in America

The 2004 GSS Finding of Shrunk Social Networks: An Artifact?

Claude S. Fischer
University of California, Berkeley

McPherson, Smith-Lovin, and Brashears (2006, 2008b) reported that Americans' social networks shrank precipitously from 1985 to 2004. When asked to list the people with whom they discussed "important matters," respondents to the 2004 General Social Survey (GSS) provided about one-third fewer names than did respondents in the 1985 survey. Critically, the percentage of respondents who provided no names at all increased from about 10 percent in 1985 to about 25 percent in 2004. The 2004 results contradict other relevant data, however, and they contain serious anomalies; this suggests that the apparently dramatic increase in social isolation is an artifact. One possible source of the artifact is the section of the 2004 interview preceding the network question; it may have been unusually taxing. Another possible source is a random technical error. With as yet no clear account for these inconsistencies and anomalies, scholars should be cautious in using the 2004 network data. Scholars and general readers alike should draw no inference from the 2004 GSS as to whether Americans' social networks changed substantially between 1985 and 2004; they probably did not.

The 2004 GSS Finding of Shrunk Social Networks: An Artifact?

Claude S. Fischer
University of California, Berkeley

McPherson, Smith-Lovin, and Brashears (2006, 2008b) reported that Americans' social networks shrank precipitously from 1985 to 2004. When asked to list the people with whom they discussed "important matters," respondents to the 2004 General Social Survey (GSS) provided about one-third fewer names than did respondents in the 1985 survey. Critically, the percentage of respondents who provided no names at all increased from about 10 percent in 1985 to about 25 percent in 2004. The 2004 results contradict other relevant data, however, and they contain serious anomalies; this suggests that the apparently dramatic increase in social isolation is an artifact. One possible source of the artifact is the section of the 2004 interview preceding the network question; it may have been unusually taxing. Another possible source is a random technical error. With as yet no clear account for these inconsistencies and anomalies, scholars should be cautious in using the 2004 network data. Scholars and general readers alike should draw no inference from the 2004 GSS as to whether Americans' social networks changed substantially between 1985 and 2004; they probably did not.

Reply to Fischer

Models and Marginals: Using Survey Evidence to Study Social Networks

Miller McPherson
Duke University

Lynn Smith-Lovin
Duke University

Matthew E. Brashears
Cornell University

Fischer (2009) argues that our estimates of confidant network size in the 2004 General Social Survey (GSS), and therefore the trend in confidant network size from 1985 to 2004, are implausible because they are (1) inconsistent with other data and (2) contain internal anomalies that call the data into question. In this note, we assess the evidence for a decrease in confidant network size from 1985 to 2004 in the GSS data. We conclude that any plausible modeling of the data shows a decided trend downward in confidant network size from 1985 to 2004. The features that Fischer calls anomalies are exactly the characteristics described by our models (Table 5) in the original article.

The 2004 GSS Finding of Shrunk Social Networks: An Artifact?

Claude S. Fischer
University of California, Berkeley

McPherson, Smith-Lovin, and Brashears (2006, 2008b) reported that Americans' social networks shrank precipitously from 1985 to 2004. When asked to list the people with whom they discussed "important matters," respondents to the 2004 General Social Survey (GSS) provided about one-third fewer names than did respondents in the 1985 survey. Critically, the percentage of respondents who provided no names at all increased from about 10 percent in 1985 to about 25 percent in 2004. The 2004 results contradict other relevant data, however, and they contain serious anomalies; this suggests that the apparently dramatic increase in social isolation is an artifact. One possible source of the artifact is the section of the 2004 interview preceding the network question; it may have been unusually taxing. Another possible source is a random technical error. With as yet no clear account for these inconsistencies and anomalies, scholars should be cautious in using the 2004 network data. Scholars and general readers alike should draw no inference from the 2004 GSS as to whether Americans' social networks changed substantially between 1985 and 2004; they probably did not.

Reply to Fischer

Models and Marginals: Using Survey Evidence to Study Social Networks

Miller McPherson
Duke University

Lynn Smith-Lovin
Duke University

Matthew E. Brashears
Cornell University

Fischer (2009) argues that our estimates of confidant network size in the 2004 General Social Survey (GSS), and therefore the trend in confidant network size from 1985 to 2004, are implausible because they are (1) inconsistent with other data and (2) contain internal anomalies that call the data into question. In this note, we assess the evidence for a decrease in confidant network size from 1985 to 2004 in the GSS data. We conclude that any plausible modeling of the data shows a decided trend downward in confidant network size from 1985 to 2004. The features that Fischer calls anomalies are exactly the characteristics described by our models (Table 5) in the original article.

Social Isolation in America: An Artifact

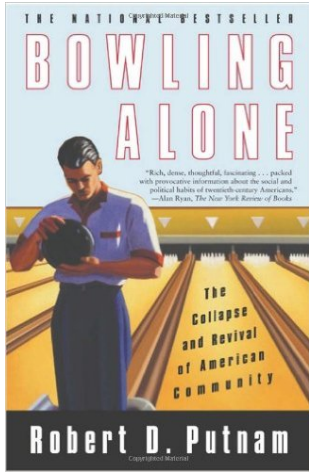
Anthony Paik^a and Kenneth Sanchagrin^a

American Sociological Review
78(2) 339–360
© American Sociological
Association 2013
DOI: 10.1177/0803322413482939
<http://asr.sagepub.com>



Abstract

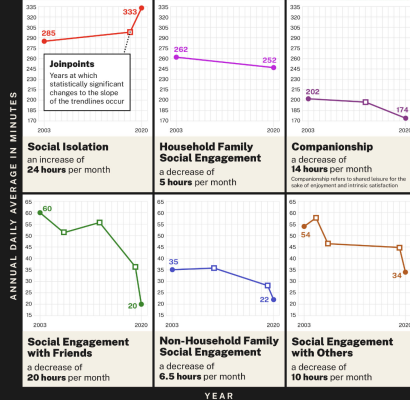
This article examines whether existing estimates of network size and social isolation, drawn from ego-centric name generators across several representative samples, suffer from systematic biases linked to interviewers. Using several analytic approaches, we find that estimates of network size found in the 2004 and 2010 General Social Surveys (GSS), as well as other representative samples, were affected by significant interviewer effects. Across these surveys, we find a negative correlation between interviewer effects and mean network size. In the 2004 GSS, levels of social connectivity are strongly linked to interviewer-level variation and reflect the fact that some interviewers obtained highly improbable levels of social isolation. In the 2010 GSS, we observe larger interviewer effects in two versions of the questionnaire in which training and fatigue effects among interviewers were more likely. Results support the argument that many estimates of social connectivity are biased by interviewer effects. Some interviewers' failure to elicit network data makes inferences, such as the argument that networks have become smaller, an artifact. Overall, this study highlights the importance of interviewer effects for network data collection and raises questions about other survey items with similar issues.



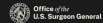
“Between 1980 and 1993 the total number of bowlers in America increased by 10 percent, while league bowling decreased by 40 percent.” Metaphor for decline of civic life.

National Trends for Social Connection

From 2003 to 2020, time spent alone increased, while time spent on in-person social engagement decreased.



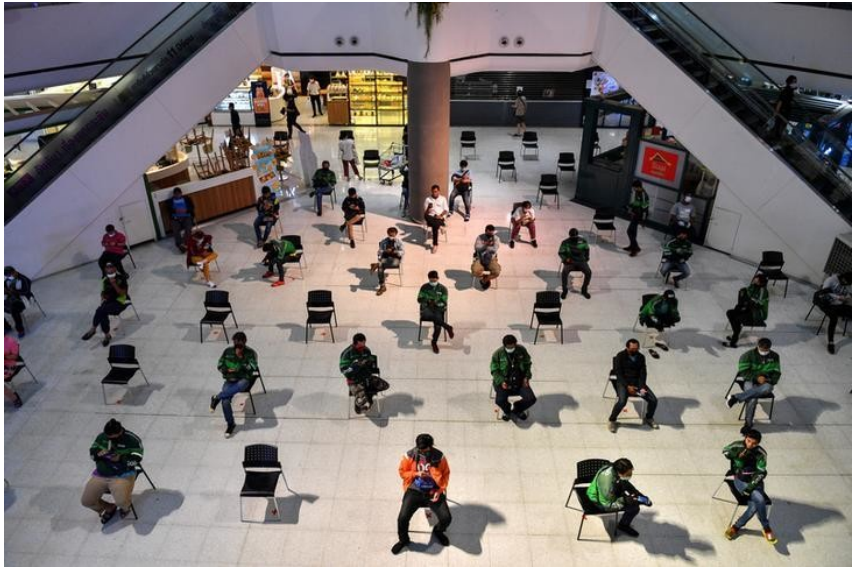
Source: Adapted from Viji Diane Kannan, Peter J. Veale, U.S. Trends in Social Isolation, Social Engagement, and Companionship: National Analysis by Age, Sex, Race/Ethnicity, Family Income, and Work Hours, 2005-2020, *SSM - Population Health*, Volume 21, 2023. The joinpoints are visual approximations.



What happened to core discussion networks during COVID?



<https://www.weforum.org/stories/2020/04/covid-19-in-pictures-this-is-what-social-distancing-looks-like/>



<https://www.weforum.org/stories/2020/04/covid-19-in-pictures-this-is-what-social-distancing-looks-like/>



<https://www.weforum.org/stories/2020/04/covid-19-in-pictures-this-is-what-social-distancing-looks-like/>

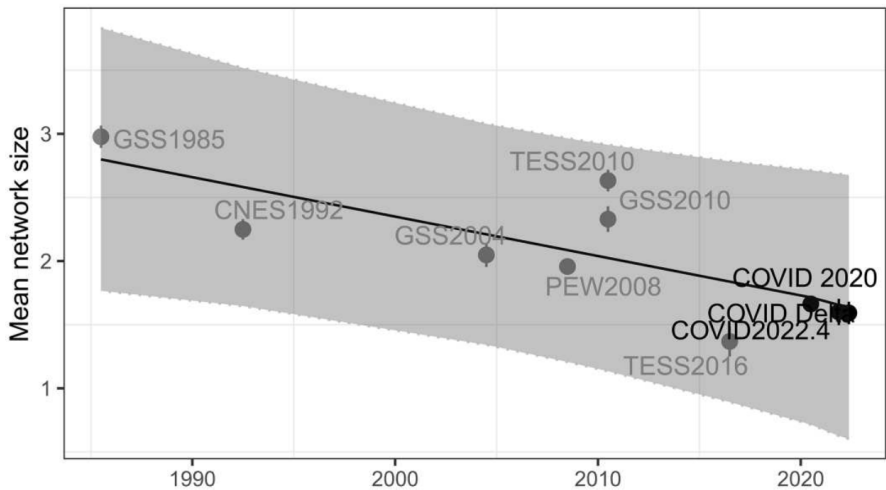


Fig. 2. The average size of core discussion networks from 1985 to 2022.

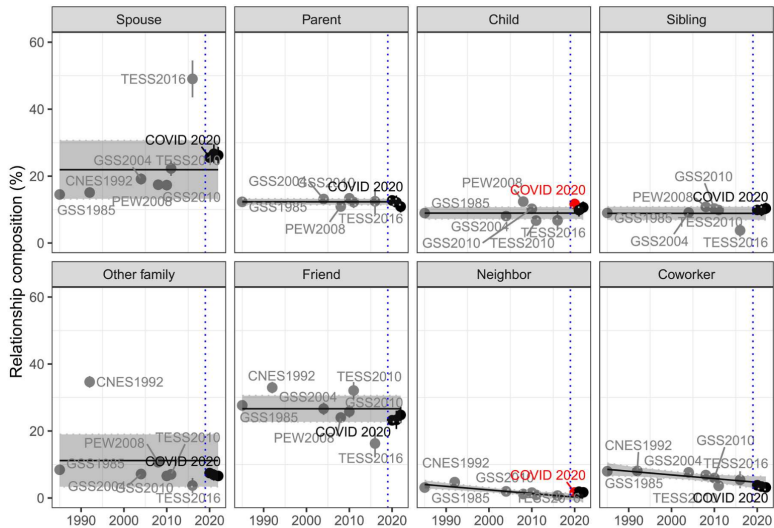


Fig. 3. The relationship compositions in core discussion networks from 1985 to 2022. Weighted means for relationship composition with 95% confidence intervals

Despite an unprecedented lockdown, we see almost no change on the long term trend in the size and composition of core discussion networks (as measured by the “discuss important matters” question).

Despite an unprecedented lockdown, we see almost no change on the long term trend in the size and composition of core discussion networks (as measured by the “discuss important matters” question).

But we do see one kind of change relative to the long term trend. ..

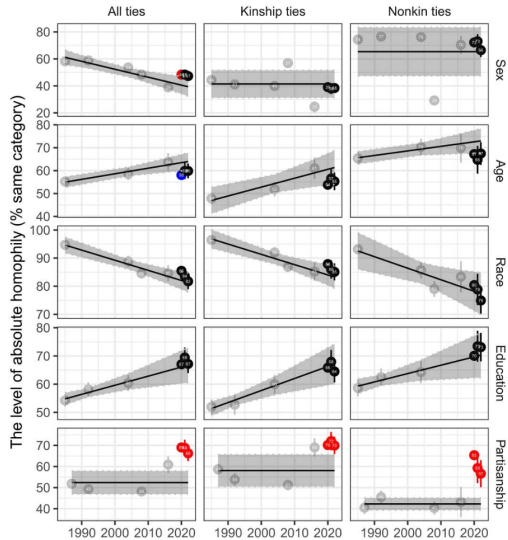


Fig. 4. The level of absolute homophily across kinship types from 1985 to 2022. Absolute homophily is measured by the proportion of alters with the

Summary:

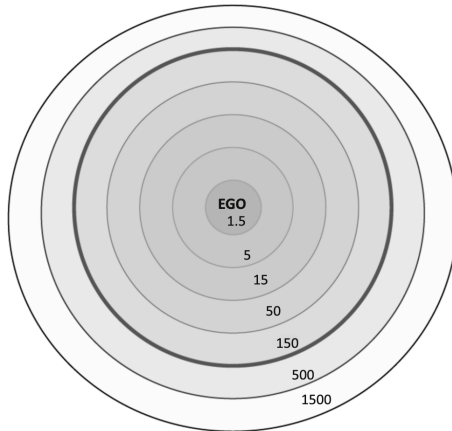
- ▶ one way to measure core discussion networks is the “discuss important matters” name generator

Summary:

- ▶ one way to measure core discussion networks is the “discuss important matters” name generator
- ▶ core discussion networks shrank from between 1985 and 2022

Summary:

- ▶ one way to measure core discussion networks is the “discuss important matters” name generator
- ▶ core discussion networks shrank from between 1985 and 2022
- ▶ core discussion networks were not effected much by COVID except an increase in homophily by partisanship



Trends in Cognitive Sciences

Figure 2. The Circles of Friendship. An individual (ego) sits within a series of hierarchically inclusive circles, or layers, of friendship that have a very distinct scaling ratio (each layer is three times the size of the one inside it). The heavy line demarcates the 150 layer, the typical size of personal social networks. Beyond this are acquaintances (500 layer); the layer at 1500 appears to represent the limit on the number of faces we can put names to. Frequency of contact, rated emotional closeness, and willingness to act altruistically towards a given alter all decline across the successive circles [2,10].

- ▶ Granovetter, M. (1973). The strength of weak ties. *American Journal of Sociology*, 78(6), 1360-1380.
- ▶ Wang and Uzzi (2022). Weak ties, failed tries, and success. *Science*, 377(6612), 1256-1258.
- ▶ Rajkumar et al. (2022). A causal test of the strength of weak ties. *Science*, 377(6612), 1304-1310.
- ▶ Singer (2022) LinkedIn Ran Social Experiments on 20 Million Users Over Five Years. *New York Times*.